2026 MCM Senior Group | Problem A

SafeInvest: Horizon-Aware Allocation with Cost-Gated Portfolio Updates

Huang Sinan, Cai Xunzhi, Lin Zihan

Fen8FenQian | 19 October 2026

Abstract

SafeInvest must recommend a portfolio a customer can live with, trade only when the gain is worth the fee, and stop advising when its own model should not be trusted. We keep these three questions separate. Three horizon-specific portfolios solve a convex problem that maximizes expected real log growth subject to a tail constraint on path log drawdown. Whether a profile may be served automatically is decided by an out-of-fold screen with a conservative binomial bound and a doubled-cost stress test. Whether a trade is worth doing is decided by a cost gate with an explicit safety margin. We call the resulting system Plan A.

All development uses 120 complete scenarios; one frozen evaluation uses the remaining 60, and only its saved outputs are reported. At 30 years and a 0.25% cost, Balanced A gives a median real annual growth of 4.15%, against 3.38% for annual equal-weight rebalancing, at a higher 90th-percentile maximum drawdown, 48.02% against 39.53%. Against annual rebalancing to the same anchor, the cost gate cuts mean real fees by 26.83% and trades at 35.34% of annual reviews. Profiles B and C receive Conservative, E receives Balanced, and A and D are referred to a human advisor. One severe path still loses 65.19% from peak to trough, so the evidence supports selective, explainable automation with escalation, not a promise of protection.

1 Decision Problem and Evidence

The advisor invests in equities, government bonds and Treasury bills. The brief joins three questions that are usually answered separately: which of the three products suits a customer, whether that customer can be served without a human, and whether an immediate trade is worth its cost. A rule that maximizes expected return answers none of them. We therefore keep the three answers apart and auditable, and we let the system decline to automate.

The supplied data are 180 independent synthetic scenarios of 30 annual periods, holding nominal returns for the three assets and inflation; the underlying series are described in the problem statement and derive from the JST Macrohistory Database, Release 6 [1, 4]. Scenario identifiers index simulations, not historical episodes. We keep each complete path as one statistical unit, since 3,600 development rows are not 3,600 independent observations, and we evaluate a customer with horizon H on the first H periods of a path.

Scenarios 1–120 form the development partition; 121–180 were held back and evaluated once, on 17 October 2026, after the model settings, the customer assignments and the comparators were frozen. That decision retains the calibrated Plan A, because later candidate families developed on the same 120 scenarios, namely prediction-tilted execution, multi-step planning, low-turnover fusion and risk-penalized value estimates, did not jointly satisfy the prespecified replacement criteria. Nothing reported from the holdout was recomputed: this report reads saved outputs, and preparing it refitted no model and re-ran no policy.

The remainder follows the three questions. Section 2 fixes the accounting and the measures of growth and loss; Section 3 builds the three portfolios and the onboarding rule; Section 4 gives the annual execution and escalation logic; Section 5 reports the frozen evaluation, Section 6 the cost stress and the limits of the evidence, and Section 7 the recommendation.

2 Accounting, Assumptions, and Performance Measures

2.1 A causal, self-contained investment account

Wealth starts at 100 units, entirely in bills. Portfolios are long-only and fully invested: $w_i \geq 0$ and $\sum_i w_i = 1$. There are no contributions, withdrawals, leverage or taxes, no fees other than the stated transaction cost, and no assumed terminal liquidation. A decision for period t uses only observations through period $t-1$ of that scenario, so every reported path is causal. Bills supply liquidity, but their real return can be negative: preservation of purchasing power is not assumed.

Let $h_{i,t}^-$ be pre-trade nominal holdings, $V_t^- = \sum_i h_{i,t}^-$, and $p_{i,t} = h_{i,t}^-/V_t^-$. A target w_t moves the following amount and incurs the following fee at rate c:

$$Q_t = \frac{V_t^-}{2}\|w_t - p_t\|_1, \qquad F_t = cQ_t, \qquad h_{i,t}^+ = (V_t^- - F_t)w_{i,t}. \tag{1}$$

The one-half factor counts each transferred amount once instead of charging the sale and the purchase separately: at $c = 0.25\%$, moving 20 units from bills to equities costs 0.05 units. Turnover is measured against pre-fee wealth, the fee is deducted before that period's returns, and the remainder is placed at the target. We state this convention rather than solve for an implicit trade that reproduces the target after the fee. The initial allocation is charged the same way, starting from an all-bill account.

For nominal return $r_{i,t}$ and inflation π_t, define real return $\rho_{i,t}$, cumulative price index I_t, and year-end real wealth W_t by

$$\rho_{i,t} = \frac{1+r_{i,t}}{1+\pi_t} - 1, \qquad I_t = \prod_{k=1}^{t}(1+\pi_k), \qquad W_t = \frac{\sum_i h_{i,t}^+(1+r_{i,t})}{I_t}. \tag{2}$$

Pending human review means existing assets stay invested and automatic trades stop; no unobserved intervention, liquidation or recovery is credited to the algorithm.

2.2 Growth, ordinary fluctuation, and unusually large loss

For a scenario with horizon H, real compound annual growth is

$$G = \left(\frac{W_H}{100}\right)^{1/H} - 1. \tag{3}$$

Maximum drawdown measures loss from a previous real-wealth peak. Let Z_j contain the initial wealth, each post-fee/pre-return wealth mark, and each year-end wealth mark. Then

$$D = \max_j \left(1 - \frac{Z_j}{\max_{k\leq j} Z_k}\right). \tag{4}$$

Including the post-fee marks keeps trading losses visible instead of hiding them between annual observations. The measure is oriented so that larger values mean a greater loss. A stated customer tolerance L is evaluated through the event $D > L$; it is a preference threshold, not a guaranteed loss limit.

The brief asks for ordinary fluctuation and unusually large loss to be separated, both in purchasing-power terms; every measure below is real. Ordinary fluctuation is the annual volatility σ, the sample standard deviation of net real annual returns. Unusually large loss is reported as the 90th percentile of D and as the 90% drawdown CVaR, which averages the worst 10% of scenario drawdowns, with fractional boundary weights where needed. CVaR is not a percentile, and the two tail figures are therefore not interchangeable. Across scenarios we report median G, P10(G), median D and P90(D), and the cross-scenario median of σ summarizes variability within a path, not dispersion across paths. Terminal-loss probability is the share of scenarios with $W_H < 100$.

The main trading frequency is the fraction of reviews in periods $2,\ldots,H$ that execute a trade, excluding the compulsory initial allocation; the rate including it is reported where relevant. Mean real fees accumulate inflation-adjusted fees over the horizon in units of initial wealth 100. Review incidence is the fraction of scenarios that ever request human review.

3 Strategic Portfolios and Customer Suitability

3.1 A convex allocation problem for each horizon

For N training paths and a target w restored every year, the largest cumulative log loss over any contiguous interval is

$$Y_s(w) = \max_{0 \le u \le v \le H} \left\{ - \sum_{t=u+1}^{v} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \right\}. \tag{5}$$

The empty interval contributes zero. On this cost-free, annually restored model an ordinary maximum drawdown equals $1 - \exp(-Y_s)$, but the expectation does not commute with that transform: $\mathrm{ES}(Y)$ does not generally give $\mathrm{ES}(D)$. We therefore constrain the quantity we can compute and report the resulting drawdown afterwards.

Using the variational representation of expected shortfall [2],

$$\mathrm{ES}_\alpha(Y) = \min_z \left[z + \frac{1}{N(1-\alpha)} \sum_{s=1}^{N} (Y_s - z)_+ \right], \quad \alpha = 0.90, \tag{6}$$

we choose the strategic anchor w^* by solving

$$\min_w \quad - \frac{1}{NH} \sum_{s=1}^{N} \sum_{t=1}^{H} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \tag{7}$$

$$\text{subject to} \quad \mathrm{ES}_{0.90}(Y(w)) \le q R_{\mathrm{ref}}, \quad w \ge 0, \quad \mathbf{1}^{\mathsf{T}} w = 1. \tag{8}$$

For each horizon, R_{ref} is the training log-drawdown expected shortfall of a stated reference allocation: $(.20, .30, .50)$ for Conservative, $(.50, .35, .15)$ for Balanced and $(.75, .20, .05)$ for Growth. The multiplier $q \in \{.75, .85, .95\}$ scales the training budget and is chosen in Section 3.2, while the validation target stays at R_{ref}. Conservative additionally requires $w_{\mathrm{bill}} \ge 0.40$. The three products thus differ both in risk budget and, for Conservative, in a liquidity floor.

Negative log growth and the maximum of interval log losses are convex on the valid-return domain, so (8) is a convex subproblem. We solve it with a Kelley cutting-plane linear program: supporting hyperplanes generate objective lower bounds, feasible weights upper bounds, and the loop stops at a gap of 10^{-6}, with risk tolerance 10^{-8} and at most 300 iterations, starting from a 231-point 5% simplex grid that also supplies a comparison objective. Primal solutions and dual multipliers are saved so that feasibility and duality balance can be rechecked. An infeasible risk budget is recorded and never repaired by quietly relaxing it. These certificates cover the strategic subproblem only; they say nothing about the optimality of the trading and escalation layer.

3.2 Development selection before freezing

Selection happens before the holdout is touched. Nested validation uses five outer folds of complete paths, 96 scenarios to train and 24 to validate, and three inner folds of 64 and 32 inside each outer training set, so every choice is scored on data the candidate was not fitted to. Candidates pair $q \in \{.75, .85, .95\}$ with execution curvature $\gamma \in \{4, 12\}$. Comparing them on held-out outcomes rather than on training fit limits the selection bias that repeated comparisons introduce [3].

A candidate is feasible on validation only if four conditions hold. Pooled log-drawdown expected shortfall at 90% must not exceed R_{ref} plus 0.005 in log units; that slack is fixed in advance and does not follow q. The median and the 10th percentile of annual real growth must not fall more than 0.001 and 0.002 below fee-paying annual rebalancing to the same anchor. Reviews, and hence the trades they permit, must be requested in at most 10% of validation scenarios. Among feasible candidates we maximize the mean of $\log(1+G)$. If none is feasible, the least-violating candidate is retained as a diagnostic and marked unusable rather than presented as a selection. All three products selected $q = 0.95$ and $\gamma = 4$ on the full development set, after which the anchors and the customer assignments were frozen.

3.3 Frozen allocations and an explicit refusal option

Table 1 gives the frozen anchors at the five horizons that appear in the questionnaire. A product name does not imply one allocation for every customer: Conservative is nearly invariant to the horizon, holding 70–74% in bills throughout, while Balanced and Growth move bills into bonds as the horizon lengthens, and the 30-year Growth solution reaches the boundary with no bills at all. The response is not monotone in the middle of the range, because each anchor is fitted separately at its own horizon. These are model outputs under the stated data and constraints, not general investment prescriptions.

Table 1: Frozen strategic equity / bond / bill weights (%) from the full-development refit; rounding may make totals differ from 100.

H	Conservative	Balanced	Growth
5	24.48 / 4.46 / 71.05	54.26 / 0.00 / 45.74	75.21 / 0.00 / 24.79
6	24.22 / 1.59 / 74.20	53.25 / 4.46 / 42.29	74.61 / 0.00 / 25.39
15	22.67 / 7.23 / 70.10	48.65 / 21.78 / 29.58	71.38 / 23.59 / 5.03
25	25.98 / 2.84 / 71.18	49.78 / 18.43 / 31.79	71.24 / 24.35 / 4.41
30	26.48 / 2.85 / 70.67	50.81 / 14.44 / 34.75	70.91 / 29.09 / 0.00

For horizon H and tolerance L we count out-of-fold paths whose complete execution policy breaches $D > L$; with k breaches among $n = 120$, the one-sided Clopper–Pearson upper bound is

$$U = \mathrm{Beta}^{-1}(1-\delta; k+1, n-k), \qquad \delta = 0.05/440, \tag{9}$$

with $U = 1$ when $k = n$. The family of 440 is fixed before screening and covers five profiles, 11 distinct product-or-programme choices, four liquidity conventions and two cost levels. Automatic service requires $U \leq 0.20$ at the base cost and at doubled cost alike. Because the out-of-fold policies are refitted on overlapping training sets, U is a conservative screening diagnostic rather than a coverage guarantee for the final refitted policy, and the 20% ceiling is a service assumption, not a stated customer preference.

Two further rules follow from the questionnaire. A high-liquidity profile requires a strategic bill weight of at least 40% in every relevant development fold; the condition is checked for any eligible product, so such a profile is not forced into the product named Conservative, and the target is not a cash floor maintained after market drift. Growth is offered only from a 15-year horizon upwards, and P90 drawdown must be ordered Conservative $\leq$ Balanced $\leq$ Growth. Among the pairs that clear every rule, the one with the highest out-of-fold median growth is recommended; if none clears them, the profile goes to human review.

Table 2 applies these rules pair by pair. For A and D no product comes close, so referral is the model's answer rather than a default. B and C clear only through Conservative, and C's Balanced alternative fails the ceiling on 17 breaches. E has two qualifying pairs; Balanced wins on out-of-fold median growth, 4.22% against 2.71%. All three served profiles also clear the doubled-cost test.

Table 2: Screening of every profile–product pair on development data: breaching paths $k/120$ and the bound U (%). Bold marks a pair that also clears the liquidity, horizon, ordering and doubled-cost rules.

Profile	H	Tol. (%)	Liquidity	Conservative		Balanced		Growth		Assigned
				$k/120$	U	$k/120$	U	$k/120$	U	
A	5	10.00	High	28/120	39.70	59/120	65.94	71/120	74.90	Human review
B	15	25.00	Medium	**3/120**	**12.45**	31/120	42.47	61/120	67.48	Conservative
C	30	40.00	Low	**0/120**	**7.29**	17/120	28.93	41/120	51.30	Conservative
D	25	15.00	Medium	58/120	65.17	97/120	91.77	107/120	96.85	Human review
E	6	35.00	High	**0/120**	**7.29**	**4/120**	**13.87**	13/120	24.69	Balanced

4 Annual Execution and Human Escalation

4.1 Candidate trades and the cost gate

At each annual review, the advisor checks for escalation before considering a trade. Let p denote current portfolio weights and w^* the frozen strategic anchor for the investment horizon. The five candidates are

$$w_\lambda = p + \lambda(w^* - p), \qquad \lambda \in \{0, .25, .50, .75, 1\}. \tag{10}$$

Here, $\lambda = 0$ keeps holdings and $\lambda = 1$ restores the anchor. Training returns supply the mean μ and regularized covariance $\Sigma = 0.75S + 0.25\,\mathrm{diag}(S) + 10^{-10}I$. The tracking score and fractional trading cost are

$$A(w) = 1 + \mu^\top w^* - \frac{\gamma}{2}(w - w^*)^\top \Sigma (w - w^*), \qquad f_\lambda = \frac{c}{2}\|w_\lambda - p\|_1. \tag{11}$$

The return term is shared across candidates; the quadratic term penalizes departure from the anchor. If any score is nonpositive or nonfinite, the advisor requests human review. Otherwise, it selects j maximizing $(1 - f_\lambda)A(w_\lambda)$ and accepts a trade only if

$$\frac{A(w_j) - A(p)}{A(w_j)} > f_j + 0.0001. \tag{12}$$

The extra 0.0001 is a one-basis-point safety margin. If the selected candidate fails this strict gate, holdings are kept; lower-ranked candidates are not tried. This score measures tracking benefit, not a guaranteed increase in future wealth.

4.2 Four observable actions

Before trading, the advisor compares observed real drawdown with the training-based threshold

$$\tau = \mathrm{clip}(1.5\ \mathrm{P90}(D_{\text{train,annual}}),\ 0.10,\ 0.75), \tag{13}$$

where $D_{\text{train,annual}}$ comes from fee-paying annual rebalancing to the same anchor. For 30-year Balanced A, $\tau = 63.85\%$. Table 3 distinguishes the four actions. Escalation takes priority over candidate selection; while review is pending, the engine holds existing assets and suspends automatic trading.

Table 3: Annual action rules. Accepted trades are classified against the previously recorded execution target, not necessarily the strategic anchor.

Action	Decision rule
1. Keep holdings	The selected step is zero or fails Equation (12); no trade fee is charged.
2. Rebalance	The accepted trade restores the recorded execution target.
3. Change target	The accepted trade sets a different execution target; w^* remains fixed.
4. Request human review	Observed drawdown reaches τ, or a candidate score is invalid; existing assets remain invested.

4.3 Customer-facing explanation

Every accepted adjustment uses this 20-word explanation, within the required 50-word limit:

> "This adjustment moves your investments toward the strategic allocation. Its estimated risk-adjusted benefit exceeds trading costs. Investment losses remain possible."

5 Final Evaluation Results

5.1 Higher growth comes with greater drawdown

Table 4 compares policies on the same 60 final scenarios, charging 0.25% of money moved. Equal-weight annual rebalancing is the required baseline; same-anchor annual and hold controls isolate execution effects.

Table 4: Thirty-year final outcomes across 60 scenarios (%). *G*: net real CAGR; *D*: maximum real drawdown. Trade rate excludes initial allocation.

Policy	Med. *G*	P10 *G*	Med. *D*	P90 *D*	Trades
Equal-weight annual	3.38	1.50	22.30	39.53	100.00
Balanced A	4.15	1.64	30.58	48.02	35.34
Same-anchor annual	4.02	1.67	30.30	48.12	100.00
Same-anchor hold	4.27	1.42	36.05	59.15	0.00
Conservative A	2.76	1.35	19.19	31.91	28.45
Growth A	5.14	2.07	38.23	62.15	28.56

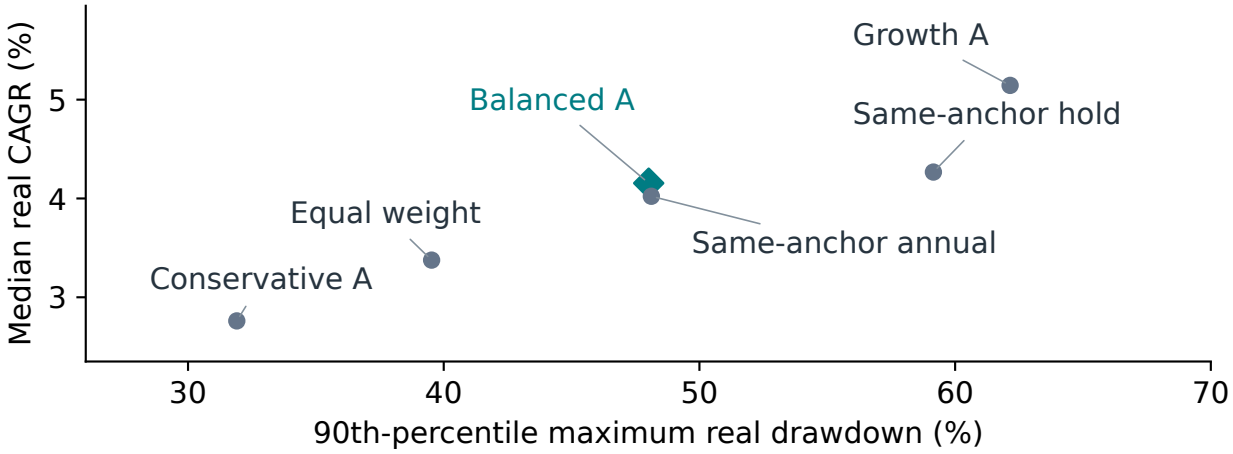


Figure 1: Thirty-year median growth versus P90 drawdown across 60 scenarios at the base fee. P90 is not a loss guarantee.

Balanced A gained 0.78 percentage points (pp) of median growth over equal weight, with 8.49 pp more P90 drawdown (Figure 1). Its median annual volatility was 10.76%, compared with 8.24% for equal weight.

Table 5: Paired bootstrap comparisons, Balanced A minus equal weight (pp); 4,000 complete-scenario resamples.

Metric	Difference	95% interval	Family-adjusted 95%
Median *G*	+0.78	[0.325, 1.184]	[0.098, 1.372]
P90 *D*	+8.49	[5.203, 13.365]	[3.399, 15.735]

Family adjustment covers 20 nonbaseline policies within each endpoint, horizon, and cost; it does not cover the whole adaptive research process.

Against same-anchor annual rebalancing, Balanced A gained 0.13 pp of median growth with nearly unchanged P90 drawdown. Mean real fees fell 26.83%, from 0.7973 to 0.5834. Its trade rate was 35.34% (37.50% including initial allocation). Relative to holding the anchor, P90 drawdown fell 11.13 pp and median growth fell 0.11 pp. These comparisons favour selective rebalancing when fees and tail losses both matter.

5.2 Customer outcomes at their actual horizons

Table 6 uses each customer's actual horizon and frozen assignment. A and D were referred for human review; dashes indicate no automated investment path. Breaches mean maximum drawdown exceeded the customer's stated tolerance.

Table 6: Final customer outcomes (growth and drawdown in %; 60 scenarios).

Customer	H	Med. G	P10 G	Med. D	P90 D	Breaches
A	5	–	–	–	–	–
B	15	2.94	0.48	12.57	25.38	7/60
C	30	2.76	1.35	19.19	31.91	3/60
D	25	–	–	–	–	–
E	6	4.11	-0.35	11.95	26.89	3/60

Table 7: Customer risk and trading. Terminal loss means final real wealth below 100; trade rates exclude initial allocation.

Customer	Median volatility (%)	Terminal losses	Trade rate (%)
B	6.26	3/60	22.86
C	6.75	3/60	28.45
E	10.08	9/60	30.67

For B, P90 drawdown exceeded the 25% tolerance, with breaches in 7 of 60 scenarios. Eligibility screens breach probability, rather than imposing a guaranteed loss limit. E's negative P10 growth and 9 terminal losses highlight the risk of its six-year horizon despite positive median growth. B, C, and E requested human review in 2, 3, and 0 scenarios, respectively. All assignments remained unchanged after final evaluation.

5.3 A severe path exposes the limits of escalation

In scenario 155, Balanced A peaked at 162.33 in year 6 and ended at 56.50 (Figure 2). Its 65.19% maximum drawdown and -1.88% real CAGR show the limits of annual escalation.

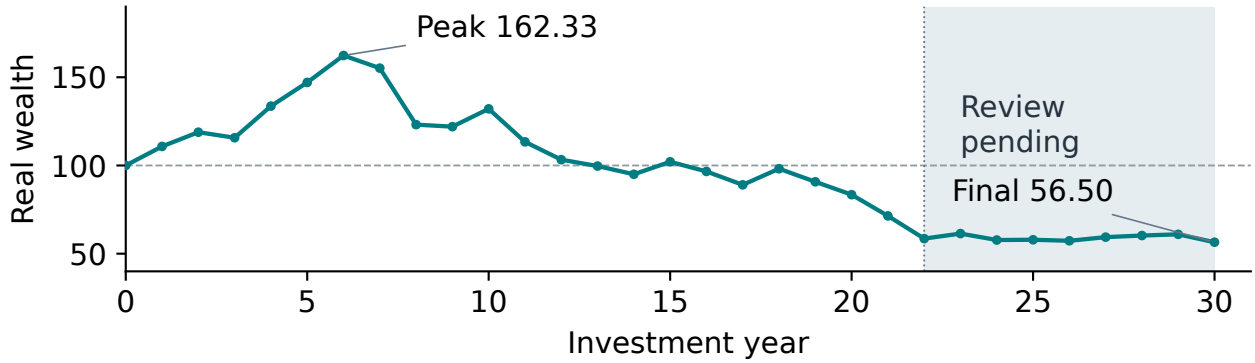


Figure 2: Saved real wealth for scenario 155, starting from 100. Lines join year-end observations; shading marks pending review, beginning before year 23.

Six post-initial trades preceded the review request before the year-23 return. Automatic trading then stopped, but the existing holdings remained exposed. Across all 60 Balanced paths, drawdown CVaR90 was 59.31%, versus P90 drawdown of 48.02%.

6 Sensitivity and Reliability

6.1 Transaction-cost stress with the policy frozen

We vary fees across five rates on the same 60 final scenarios, retaining the frozen strategic parameters and customer assignments. The cost gate responds to each rate (Table 8, Figure 3).

Table 8: Thirty-year Balanced A, 60 final scenarios. Growth and drawdown are in %; mean fees are cumulative real wealth units (initial wealth 100). Trade rates exclude initial allocation.

Cost (%)	Med. G	P10 G	P90 D	Mean fees	Trades (%)
0.000	4.08	1.68	47.96	0.0000	46.61
0.125	4.19	1.70	47.56	0.3199	39.60
0.250	4.15	1.64	48.02	0.5834	35.34
0.500	4.13	1.61	47.35	1.0668	28.68
1.000	4.13	1.61	47.72	1.8687	19.83

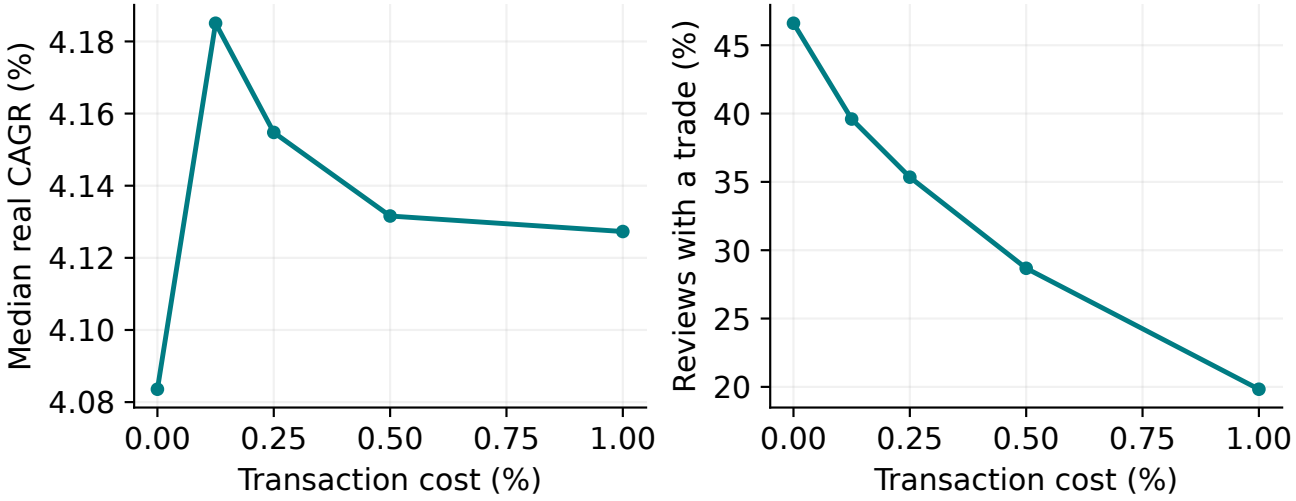


Figure 3: Trading decreases across the tested fee rates. Growth is nonmonotonic because changing fees also changes the trades.

Doubling the fee to 0.50% reduces trading from 35.34% to 28.68%; median growth changes from 4.15% to 4.13%, and mean real fees rise from 0.5834 to 1.0668. Nonmonotonic growth reflects changed exposures, not a benefit from fees.

For eligible customers, selective Balanced updates remain our recommendation: at every tested rate, median growth and P90 drawdown exceed equal weight. Customer eligibility was screened only at base and doubled costs during development.

6.2 What is verified, and what remains uncertain

Saved checks covered 286,920 transitions, 780 equal-weight scalar-reference cases, and five customer assignments. Maximum absolute reconstructed year-end holding error was 3.64×10^{-12} nominal wealth units. These checks support accounting consistency, not predictive reliability.

Synthetic data and only 60 independent final paths limit generalization and tail-risk precision; testing five fees does not add independent paths. Annual reviews miss within-year drawdowns, and the cost-free strategic risk proxy differs from realized execution risk. Bill weights can drift below targets. Withdrawals, taxes, additional frictions, and effective human intervention are unmodeled.

7 Recommendation and Conclusions

We recommend the frozen calibrated Plan A. Conservative uses the lowest reference risk budget and a strategic bill floor of 40%; Balanced uses an intermediate budget; Growth uses the highest budget and is eligible only for horizons of at least 15 years. Their horizon-specific allocations are given in Table 1. Applying the suitability screen at each customer's horizon assigns B and C to Conservative, E to Balanced, and A and D to human review.

Balanced should use selective automated updates when estimated benefit exceeds transaction costs and the safety margin. At 30 years and the base fee, this cuts mean real fees by 26.83% relative to annual rebalancing to the same anchor, with similar P90 drawdown. Against equal weight, median growth is higher but tail drawdown is worse. This supports cost-aware execution for suitable customers, without establishing universal risk or return superiority.

The main safeguard combines refusal at onboarding with suspension of automatic trades when observed real drawdown reaches the training-derived review threshold or model scores are invalid. Existing assets remain invested pending human review. For 30-year Balanced, the threshold is 63.85%, and the severe path still loses 65.19% from peak to trough. Escalation therefore cannot be presented as capital protection.

Reproducibility and attribution

The frozen model, development selection record, one-time final-evaluation lock, and comparator outputs preserve the evaluation trail. Report tables and figures use these saved records. Supplementary source code is submitted separately.

The problem and supplied synthetic data are credited to the 2026 MCM Senior Group organizers and the stated underlying JST source [1, 4]. We acknowledge their CC BY-NC-SA 4.0 license; derived data displays retain this attribution and the same terms. Reported returns describe the synthetic experiment and are not forecasts.

References

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Use of AI Tools

OpenAI ChatGPT/Codex assisted with code development, analysis, drafting, and document checks. Numerical claims in this report are drawn from saved experiment outputs and frozen model settings; the report-building step did not refit a model or rerun the final evaluation. AI-generated explanations and prose require review by the named team members before submission. The authors remain responsible for the correctness, originality, attribution, and final submitted content.